Methodology + Experimental Design

Excerpt from "Soft vs Rigid Gripper Benchmark for Cloth Grasping"

A thesis in progress, McGill University — for supervisor review

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Compiled August 7, 2026

This standalone document contains only Chapters 3 (Methodology) and 4 (Experimental Design and Procedure) of the thesis manuscript. Chapter and section numbering starts fresh at 1 here; the equivalent chapter numbers in the full thesis are noted in the running header.

Contents

Notation

Frames and poses

$\mathcal{W}$, $\mathcal{F}$, $\mathcal{T}$	World (robot base), gripper flange, and gripper tip frames
$\mathbf{p}$	Position vector in $\mathbb{R}^3$ (boldface); $\mathbf{p}_{\mathcal{F}}$, $\mathbf{p}_{\mathcal{T}}$ denote the flange and tip positions expressed in $\mathcal{W}$
R, R_0	Orientation of the gripper as a rotation matrix, $R \in SO(3)$; $R_0 = R_x(180°)$ is the nominal down-pointing orientation
$R_x(\phi)$, $R_y(\phi)$	Elementary rotation matrix for a rotation by angle ϕ about the fixed world x or y axis
$\hat{\mathbf{z}}$	Unit vector along $+z$
z_{TCP}	Per-variant tool-centre-point offset from flange to tip along the flange's local z axis (mm); $z_{\mathrm{TCP}}^{(v)}$ denotes the calibrated value for variant v
x_0, y_0	Fixed probe/target position in the world xy plane (mm)

Design parameters (fixed per variant)

t	Finray finger blade thickness (mm)
α	Baseline seat-mount tilt angle, about the gripper's local y axis, introduced by the 3D-printed inclined seat (°)

Runtime and trial variables

θ_x, θ_y	Commanded robot-arm tilt angle about the world x / y axis (which coincide with the gripper's local x / y axis lines at R_0), applied as $R_{\text{tilt}} = R_y(\theta_y)\, R_x(\theta_x)$ (extrinsic) (°); distinct from, and never summed with, the seat angle α
d	Press depth: distance the gripper fingertip is driven below the detected cloth/table surface (mm)
z_{surf}	Detected cloth/table surface height at the tip's xy location (mm)
a	Tilt axis identifier (x or y)
$b(d, a)$	Signed tilt bias: $b(d, a) = \lvert F_z^{+2.5°}(d, a)\rvert - \lvert F_z^{-2.5°}(d, a)\rvert$ (N)

Force measurement

$\mathbf{F}_{\text{ext}}$	Franka external wrench force vector at the end-effector flange (N); F_x, F_y, F_z denote its components
F_z^{bias}	Free-air F_z bias captured at session start (N)
$F_z^{(i)}$	The i-th stationary F_z sample within a trial (N)
N	Number of stationary force samples pooled in a (variant, depth, tilt) cell
$\bar{F}_z$, σ_{F_z}	Per-cell mean and standard deviation of F_z over pooled trials
$\sigma_{z,\text{tcp}}$	Across-tap standard deviation of the touch-off z measurement during TCP calibration (mm; distinct from σ_{F_z} above)

Derived quantities

L_{eff}	Effective finger lever-arm length inferred from the mount-tilt/depth-shift equivalence (mm)

Chapter 1

Experimental Setup and Methodology

1.1 Hardware Configuration

1.1.1 Robot Manipulator

A Franka Emika FR3 7-DOF cobot served as the manipulator throughout this work. The arm was operated through the Franka Control Interface (FCI) using `libfranka 0.18.1` and the ROS 2 Humble control stack. Cartesian pose tracking was performed by a custom `fr3_pose_controller` maintained by the McGill Applied Dynamics Lab at `https://github.com/McGill-Applied-Dynamics-Lab/adl-ros2`, parameterised in `config/controllers/fr3_pose/default.yaml`. The controller accepts a sequence of (pose, twist) waypoints with associated time stamps and interpolates between them using a quintic polynomial; for the in-place re-orientation and the final precision descent phases, a cosine half-cycle S-curve interpolator (the so-called "sine descent" implemented in `run_trial.py::_safe_traj`) is used to ensure zero start- and end-state velocities.

1.1.2 Gripper Variants Tested

The finray fingers and the parallel-jaw compliant pad were 3D-printed from Bambu Lab TPU 90A (Shore-A hardness 90) on a Bambu Lab X1 Carbon printer using the vendor-default TPU 90A profile in Bambu Studio (0.4 mm nozzle, default layer height, default infill, PEI-textured build plate).

Table 1.1: Six gripper variants benchmarked, indexed by blade thickness t and baseline seat-mount tilt α (Figure 1.1). All finray variants share the same 3D-printed inclined-seat mount; only α changes between the $+2.5°$ baseline and the $+5.0°$ variant. The parallel-jaw variants do not use an inclined seat and have no defined t or α.

Variant	Type	t	α
`parallel_jaw_stock`	Rigid two-finger (Franka Hand)	—	—
`parallel_jaw_TPU`	Rigid two-finger + 2.0 mm TPU 90A pad	—	—
`finray_18`	Finray-effect 3D-printed	18 mm	$+2.5°$
`finray_18_model2`	Second print of `finray_18`	18 mm	$+2.5°$
`finray_26`	Finray-effect 3D-printed	26 mm	$+2.5°$
`finray_26_5deg`	Finray-26 on 5°-seat mount	26 mm	$+5.0°$

Baseline inclined-seat mount. We define α as the baseline seat-mount tilt angle: the fixed rotation about the gripper's local y axis (the gripping direction) introduced by the 3D-printed seat adapter that carries every finray finger. All finray variants were mounted on the Franka Hand via this inclined seat; the seat was present in *every* finray variant tested in this thesis unless otherwise noted, and is the physical realisation of α. Variants whose name contains `_5deg` used a $\alpha = +5.0°$ seat in place of the default $\alpha = +2.5°$ seat; all other finray variants (`finray_18`, `finray_18_model2`, `finray_26`) used the $\alpha = +2.5°$ baseline; the parallel-jaw variants have no seat and no defined α. The $\alpha = +2.5°$ baseline was chosen so that the two opposing finray fingers converge inward toward each other rather than splaying apart, closing the residual slit that would otherwise remain between the finger tips when the gripper is fully closed, ensuring firm contact between the fingers along their full length during a successful grasp. Figure 1.1 shows the mounting stack (flange $\rightarrow$ seat $\rightarrow$ finger) and defines the α and t design-parameter convention.

Consequence for reported "0°" cells. Cells reported in Chapter Results ch. (full thesis) as 0° tilt mean the robot arm is perpendicular to the surface ($\theta_x = \theta_y = 0$). Runtime tilt values ($\pm 2.5°$ about x or y) tilt the arm away from this perpendicular pose.

1.1.3 Cloth Specimen

[TODO: insert photograph of the cloth specimen and short caption describing material, thickness, weave, and dimensions.]

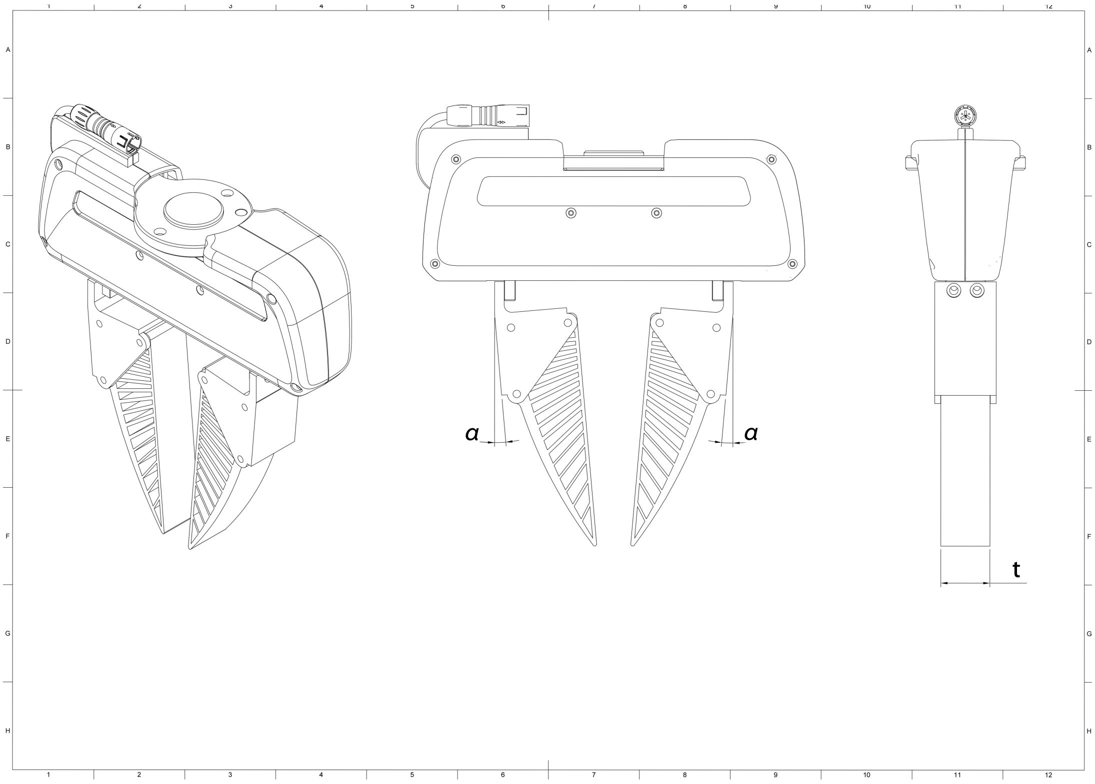


Figure 1.1: Isometric, front, and side orthographic views of a finray variant mounted on the Franka Hand. The front view (centre) annotates the baseline seat-mount tilt α at each finger root; the side view (right) annotates the finger blade thickness t. **[TODO: add explicit gripper-frame axis labels (x, y, z) to each view once the frame-triad figure is finalised, per supervisor request.]**

1.1.4 Test Surface

[TODO: insert photograph of the test bench in situ and short caption describing the table material, flatness, and cloth-edge orientation relative to the gripper y axis.]

1.2 Reference Frame and Tool-Centre Calibration

1.2.1 Frame Convention

Three frames are relevant: the world (robot base) frame $\mathcal{W}$, the gripper flange frame $\mathcal{F}$, and the gripper tip frame $\mathcal{T}$. The Franka URDF defines $\mathcal{F}$ at the mechanical hand mounting flange; the Franka API `robot.end_effector_pose` reports the pose of $\mathcal{F}$ in $\mathcal{W}$. The tip frame $\mathcal{T}$ is located along the local $+z$ axis of $\mathcal{F}$ by a per-variant offset z_{TCP}, which depends on the physical gripper (finger length, mounting fixtures, pads).

Throughout, $\mathbf{p} \in \mathbb{R}^3$ (boldface) denotes a position vector expressed in $\mathcal{W}$ unless otherwise noted, and $R_x(\phi)$, $R_y(\phi)$ denote the elementary rotation matrices for a right-handed rotation by angle ϕ about the fixed world x or y axis, so that a general gripper orientation is a composition $R \in SO(3)$ of these primitives. For a gripper at orientation R, the world-frame relationship between the tip and flange positions is

$$\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} + R \begin{bmatrix} 0 \\ 0 \\ z_{\mathrm{TCP}} \end{bmatrix}. \tag{1.1}$$

For the rigid-down-pointing nominal orientation $R_0 = R_x(180°)$, the tip lies directly below the flange and $\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} - z_{\mathrm{TCP}}\hat{\mathbf{z}}$.

The gripper-frame convention adopted here is: the gripping axis (along which the fingers close) is $\mathcal{F}$'s local y axis; the gripper is mirror-symmetric about the xz plane; consequently, rotation of the gripper about the local y axis is rotation about the geometrically asymmetric axis, while rotation about x is rotation about the symmetric axis. Because the nominal orientation $R_0 = R_x(180°)$ is itself a rotation about the world x axis, the world x and y axis *lines* coincide with the flange's local symmetric and gripping axis lines at R_0 (up to sign). The runtime tilt angles θ_x, θ_y introduced in Section 1.4 are defined as rotations about these fixed world axes, and because the axis lines coincide with the gripper-local axes at R_0, the same symmetric-axis / gripping-axis language applies without ambiguity to θ_x and θ_y respectively.

1.2.2 TCP Offset Calibration

The z_{TCP} value differs by variant because different fingers attach to the same flange. To calibrate, each variant was driven to a common probe position (x_0, y_0) on a known rigid table and a force-feedback touch-off was performed: the gripper descended slowly in z until the external force in z exceeded a $-3\,\mathrm{N}$ threshold (debounced for three consecutive $50\,\mathrm{Hz}$ samples). The flange z at the latch moment was recorded as $z_{\mathrm{TCP}}^{\mathrm{measured}}$.

Taking the rigid `parallel_jaw_stock` variant as the URDF reference ($z_{\mathrm{TCP}} \equiv 0$ by definition), the per-variant TCP offset is

$$z_{\mathrm{TCP}}^{(v)} = z_{\mathrm{TCP}}^{(v),\mathrm{measured}} - z_{\mathrm{TCP}}^{\mathrm{stock,measured}}. \tag{1.2}$$

Table 1.2 summarises the calibrated values used throughout this work.

Table 1.2: Calibrated per-variant TCP offsets from differential touch-off against the stock parallel jaw.

Variant	$z_{\mathrm{TCP}}^{\mathrm{measured}}$ (mm)	$z_{\mathrm{TCP}}^{(v)}$ (mm)
`parallel_jaw_stock`	37.871	0.000
`parallel_jaw_TPU`	39.916	2.044
`finray_18`	131.842	93.971
`finray_26`	132.782	94.911
`finray_26_5deg`	130.094	92.223

The reproducibility of the touch-off was measured as the across-tap standard deviation $\sigma_{z,\mathrm{tcp}}$ in mm (not to be confused with the per-cell force standard deviation σ_{F_z} of Section 1.7). For `finray_26_5deg`, $\sigma_{z,\mathrm{tcp}} = 183\,\mu\mathrm{m}$ owing to a wobble component introduced by the inclined seat; the other four variants had $\sigma_{z,\mathrm{tcp}} \leq 102\,\mu\mathrm{m}$.

1.3 Force Measurement

External forces $\mathbf{F}_{\mathrm{ext}}$ at the flange were estimated by the Franka internal model from joint torques, available via the `libfranka` RobotState API at 1 kHz and subsampled to 30 Hz for session logging. A free-air bias capture was performed at the start of every session: the gripper was held at a known pose 30–80 mm above the cloth surface and 30 samples were averaged over 1 second to provide F_z^{bias}. This bias was recorded in each session JSON but *not* subtracted from the raw force time series at acquisition time; downstream analyses subtract it where appropriate.

1.4 Tilt-Aware Pose Targeting

When the gripper is tilted, the flange position required to place the tip at a fixed probe location $(x_0, y_0, z_{\mathrm{surf}})$ shifts laterally. From equation (1.1), the required flange position is

$$\mathbf{p}_{\mathcal{F}}^{*} = \mathbf{p}_{\mathcal{T}}^{\mathrm{target}} - R \begin{bmatrix} 0 \\ 0 \\ z_{\mathrm{TCP}} \end{bmatrix}. \tag{1.3}$$

This relation is implemented in the trial controller `run_trial.py`: the target orientation $R = R_{\mathrm{tilt}} \circ R_0$ is constructed at session start by composing the requested tilt rotation $R_{\mathrm{tilt}} = R_y(\theta_y) R_x(\theta_x)$ (extrinsic, world frame) onto the base orientation $R_0 = R_x(180^\circ)$, and the flange position target is updated accordingly. The orientation is then held constant through all subsequent descent and grasp phases, so the tip XY remains at (x_0, y_0) throughout the trial. Figure 1.2 illustrates the two runtime tilt angles: θ_x rotates the gripper about the symmetric axis (front view, both fingers move together), while θ_y rotates it about the gripping axis (side view, the mirror-symmetric fingers overlap in projection). Both are shown here as a general rotation from the world z axis ("Ground"); the runtime sweep used in this study is $\theta_x, \theta_y \in \{-2.5°, 0°, +2.5°\}$. These arm rotation angles are a separate quantity from the seat angle α of Section 1.1.2: α is a fixed property of the gripper hardware (how the fingers meet the flange), while θ_x, θ_y describe how the arm is oriented relative to the surface. The two are reported independently throughout this thesis and are never summed.

For a $+2.5°$ tilt about x on the `finray_26` variant ($z_{\mathrm{TCP}} = 94.911\,\mathrm{mm}$), the predicted flange offset is

$$\Delta y_{\mathcal{F}} = -z_{\mathrm{TCP}} \sin(2.5°) \approx -4.14\,\mathrm{mm}. \tag{1.4}$$

Chapter 2 reports the empirical lateral offsets measured under load and the residuals from this geometric prediction, which we attribute to finger compliance under press.

1.5 Data Pipeline

1.5.1 Per-Trial Recording

Each trial directory `trial_NNN/` contains:

- `config.yaml` — snapshot of the variant config at trial time;
- `run_config.json` — depth, tilt, ground-truth source, speeds, gripper mode;
- `pose_wrench.csv` — high-rate (30 Hz) actual + commanded pose, RPY, wrench, gripper width per row, tagged with the current trial stage.

1.5.2 Per-Session Recording

Each session directory `depth_*/` contains in addition:

- `depth_summary.json` — header metadata + array of per-trial verdicts (`success`, `fail_reason`); operator notes;
- `ground_truth_used.json` — snapshot of the GT JSON loaded;

- `session_log_*.npz` — numpy archive of the 30 Hz background sample stream, with stage labels and tap indices.

1.5.3 Verdict Recording and Outcome Classification

Trial outcomes were classified into three categories rather than a binary success/failure. After each trial the operator was prompted to distinguish *success* (the gripper held the cloth throughout the lift stage) from *no-grasp* (the trial executed to completion and the stationary force sample was valid, but the cloth slipped from the gripper during or after the lift). Trials in which the FCI control daemon aborted the motion mid-trial were recorded as *failed trials* and excluded from all aggregate analyses; the two most common triggers were:

- the Franka safety reflex firing on excessive predicted joint force (`auto_failed_excess_force` which typically occurred when the parallel-jaw variants were commanded to depths beyond their narrow force-limited working range;
- the Franka safety reflex firing on acceleration discontinuity between two consecutive Cartesian trajectory segments, which was observed intermittently during the chunked touch-off descent and mitigated through the trajectory-emission-floor modifications documented in Section 1.2.2.

Both trigger classes left the trial's `success` field as `null` in the per-trial JSON; no-grasp trials, by contrast, carried `success = false` with a valid force record and were retained for engagement-force analysis.

1.6 Trial Procedure

A single trial proceeded as follows:

1. The gripper opened (Franka Hand or finray release width);
2. Bulk descent: a smooth multi-waypoint trajectory carried the gripper from the lifted pose down to a fine clearance (+5 mm above surface), filtered to exclude any upward leg (preventing direction reversals at zero-twist interior waypoints);
3. Settle: a 2 s pause at the fine-clearance pose;
4. Final approach: a cosine-S-curve descent carried the gripper from the fine clearance to the press target $z_{\text{surf}} - d$, where d is the trial press depth (Figure 1.3);

5. Grasp: a force-controlled close (Franka grasp action at 80 N or 20 N nominal, with ±100 mm ϵ to disable the position-acceptance check that would otherwise cause auto-release on a sub-spec final width);

6. Settle: a 1 s pause;

7. Stationary sample: 30 samples of $\mathbf{F}_{\text{ext}}$ were captured over 1 s during the `grasp_stationary_s` stage;

8. The gripper lifted to $\sim +80$ mm above surface for visual inspection;

9. The operator was prompted for a verdict;

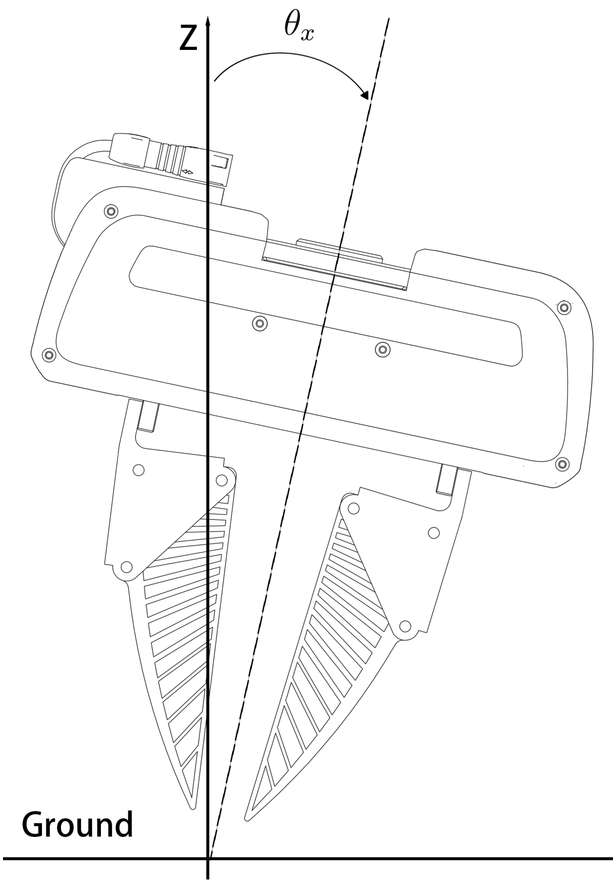
10. The gripper opened and the rig was reset for the next trial.

1.7 Statistical Treatment

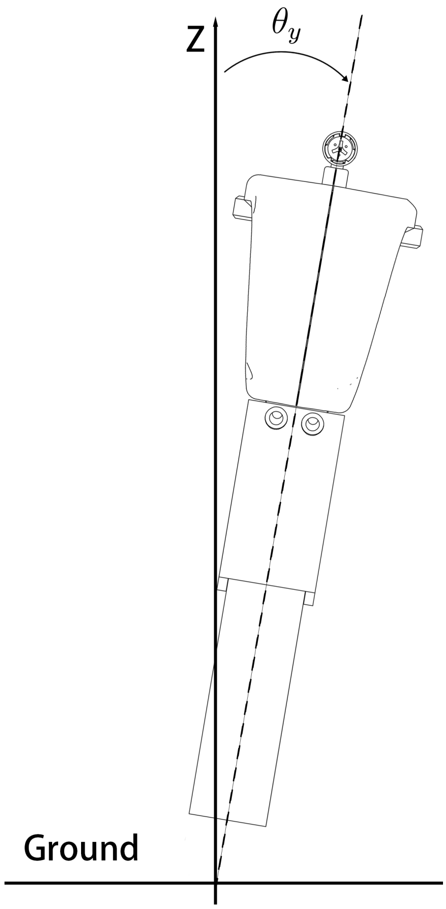
For each (variant, depth, tilt) cell, the contact force is reported as

$$\bar{F}_z = \frac{1}{N}\sum_{i=1}^{N} F_z^{(i)}, \qquad \sigma_{F_z} = \sqrt{\frac{1}{N}\sum (F_z^{(i)} - \bar{F}_z)^2}, \tag{1.5}$$

where N is the total number of stationary samples pooled across all clean trials in the cell. Trials with null verdict (FCI crash) are excluded. Trial-to-trial variance is the dominant source of dispersion; the within-trial sample variance (~ 0.1 N) is small compared to it.



(a) θ_x: rotation about the symmetric axis (front view)



(b) θ_y: rotation about the gripping axis (side view)

Figure 1.2: Runtime tilt angles θ_x (left) and θ_y (right), each shown as a rotation of the whole gripper body away from the world z axis relative to Ground. **[TODO: add explicit $x/y/z$ axis labels and the $\mathcal{F}/\mathcal{T}$ frame origins to both panels to close out the full frame-triad request.]**

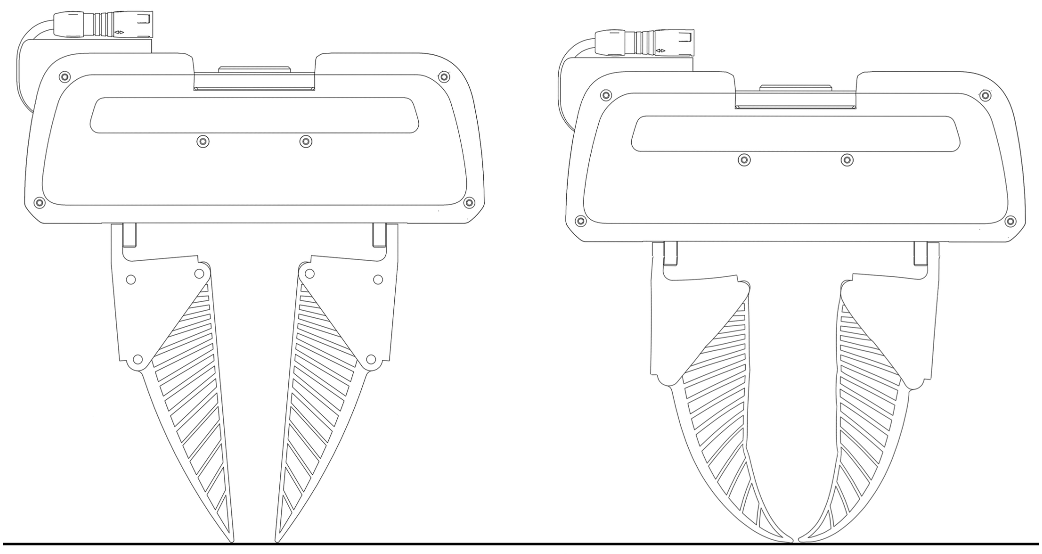


Figure 1.3: Physical effect of the press depth d on a finray variant. At first contact ($d = 0$, left) the blades are straight and only the tips touch the surface; pressed to $d > 0$ (right), the compliant blades buckle outward and wrap the surface, increasing the contact area. This wrap-around engagement underlies the depth-gated tilt robustness discussed in Chapter Discussion ch. (full thesis).

Chapter 2

Experimental Design and Procedure

The empirical campaign is organised into two sequential phases against a single experimental apparatus. Phase 1 establishes the contact-force versus press-depth relationship for each gripper variant at nominal zero-tilt orientation, and phase 2 perturbs the same measurement with $\pm 2.5^\circ$ tilts about the two orthogonal Cartesian axes. Together the two phases define a three-dimensional (depth, tilt-axis, tilt-sign) cell grid whose per-cell measurements constitute the data corpus reported in Chapter Results ch. (full thesis).

2.1 Study Design

The gripper roster comprises six variants organised into two families:

- **Finray family** (four variants): `finray_18`, `finray_18_model2` (second print of the same CAD), `finray_26` (larger blade), and `finray_26_5deg` (`finray_26` on a 5°-inclined seat).
- **Parallel-jaw family** (two variants): `parallel_jaw_stock` (bare rigid jaws) and `parallel_jaw_TPU` (rigid jaws with a 2 mm TPU 90A compliant pad).

The same fabric specimen (single-layer cotton) rests on the same acrylic table surface throughout the campaign. The Franka Emika Panda arm holds the gripper in a fixed base orientation $R_0 = R_x(180^\circ)$ for phase 1, and applies signed rotations $R_{\alpha,\beta} = R_x(180^\circ)R_x(\alpha)R_y(\beta)$ for phase 2, with $(\alpha, \beta) \in \{(\pm 2.5^\circ, 0), (0, \pm 2.5^\circ)\}$.

2.2 Depth Grid

The depth axis is defined relative to the per-session detected cloth surface via the `ground_truth_finder` touch-off procedure described in Chapter 1. The depth grid was chosen adaptively per gripper family:

- For the rigid parallel jaws, depths are spaced at 0.25–0.5 mm intervals over −1 to +4 mm, close to the narrow force-limited working range.
- For the finrays, depths are spaced at 0.5–1.0 mm intervals over −2 to +16 mm, covering the four regimes (boundary, knee, valley, recovery) identified in Chapter Results ch. (full thesis).

Per (variant, depth, tilt) cell, three trials were collected. Boundary depths were routinely retested when within-cell standard deviation exceeded 1 N, up to a maximum of two additional cells per (depth, tilt) combination.

2.3 Tilt-Aware Pose Geometry: Verification

Before reporting force results, the implementation of the tilt-aware tool-centre-point (TCP) tracking equation was verified empirically. Table 2.1 shows the measured flange XY shift during the stationary grasp sample versus the geometric prediction for `finray_26` at four representative (depth, tilt) cells.

Table 2.1: Measured vs. predicted flange-XY offset during the stationary grasp sample at four cells. Predicted offset uses $z_{\text{TCP}} = 94.911$ mm.

Cell	Tilt cmd	Predicted $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Measured $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Residual	Mean $\lvert\mathbf{F}\rvert$
−1.0 mm	+2.5° y	(+4.14, +0.00)	(+4.35, −0.03)	0.21 mm	−8.00 N
−1.0 mm	−2.5° y	(−4.14, +0.00)	(−4.02, −0.04)	0.13 mm	−3.39 N
+10.0 mm	+2.5° y	(+4.14, +0.00)	(+5.35, −0.28)	1.25 mm	−36.26 N
+10.0 mm	−2.5° y	(−4.14, +0.00)	(−2.93, −0.29)	1.21 mm	−35.95 N

At boundary depth, the residual is ≤ 0.21 mm—essentially the lateral positioning noise of the arm under light load. At safe depth, both positive and negative tilts exhibit a +1.2 mm common-mode offset in the same $+x$ direction, regardless of tilt sign. We interpret this common-mode offset as evidence of finger compliance under load: the ~ 36 N axial press deflects the soft finray blade laterally by approximately 1.2 mm in a structurally preferred direction. The differential offset (between $+y$ and $-y$ tilts) remains correctly predicted by the geometric model in both depth conditions.

This verification confirms that the control-frame implementation of the tilt-aware TCP tracker is correct, and any asymmetries observed in the contact force reflect physical gripper behaviour rather than control-system errors.

2.4 Rotation Tracking Verification

A natural concern is whether the Franka arm faithfully realises the commanded tilt orientation. The 30 Hz session log records actual and commanded RPY at every sample, enabling a direct measurement of orientation tracking error

$$\delta(t) = \|\mathrm{RPY}_{\mathrm{actual}}(t) - \mathrm{RPY}_{\mathrm{cmd}}(t)\|_2. \tag{2.1}$$

Across five representative sessions, the mean steady-state δ was 0.05° with a worst-case 0.13°. Transient excursions up to $\sim 5°$ occur exclusively within the `orientation_lock` stage at session start, before any descent or grasp, and are filtered out of trial-level analyses.

A 2.5° commanded tilt is therefore 50× the noise floor of the arm's rotation tracking; the observed force differences cannot be attributed to arm rotation error.

2.5 Trial Protocol and Outcome Classification

Each trial follows a fixed six-stage sequence: (1) set the commanded pose above the target, (2) descend to the pre-contact height, (3) contact and press to the target depth, (4) grasp with the gripper's default close command, (5) lift by a fixed clearance, (6) operator confirms whether the cloth is held. A stationary-sample window of at least 2 seconds during the grasped-and-pressed stage provides the force measurement.

Trial outcomes are classified into three categories:

Success: the trial executed to completion, contact force was sampled cleanly during the stationary window, and the cloth remained gripped after the lift stage.

No-grasp: the trial executed to completion and contact force was sampled cleanly during the stationary window, but the cloth slipped from the fingers during or after the lift stage. The force measurement remains valid as an *engagement-force* characterisation.

Failed trial: the trial did not execute to completion. The controller either aborted with an excess-force safety event or the operator interrupted before the stationary window completed. The measurement is invalid and the trial is excluded from analyses.

The rationale for this three-way taxonomy—rather than a binary success/failure classification—is discussed in Section Failure-Mode Discussion (full thesis).